

Flow Transformer: A Novel Anonymity Network Traffic Classifier with Attention Mechanism

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Abstract—Supervising anonymity network is a critical issue in the field of network security, and traditional traffic analysis methods cannot cope with complex anonymity traffic. In recent years, the traffic analysis method based on deep learning has achieved good performance. However, most of the existing studies do not consider the temporal-spatial correlation of the traffic, and only use a single flow for classification. A few works take continuous flows as flow sequence for traffic classification, but they do not distinguish the different importance of each flow. To tackle this issue, we propose a novel flow-based traffic classifier called FLOW TRANSFORMER to classify anonymity network traffic. FLOW TRANSFORMER uses multi-head attention mechanism to set higher weights for important flows, and extracts flow sequence features according to the importance weights. Besides, the RF-based feature selection method is designed to select the optimal feature combination, which can effectively avoid the insignificant features from reducing the performance and efficiency of the classifier. Experimental results on two real-world traffic datasets demonstrate that the proposed method outperforms state-of-the-art methods with a large margin.

Index Terms—Anonymity network, traffic classification, deep learning, transformer, attention mechanism.

I. INTRODUCTION

With the continuous development of Internet services, people pay more attention to the privacy and security of the Internet, which can be well met by anonymity networks. In recent years, anonymity communication technology has developed rapidly, and onion routing theory based on the MIX network has been widely used. Many anonymity networks (e.g. Tor¹, I2P², JonDoNym³, etc.) are implemented based on the MIX network. Based on the different anonymity characteristics of these networks, a variety of technologies (e.g. content encryption, multi-hop mechanism, traffic obfuscation) are used to realize the hiding of communication relationships and content. At present, the number of anonymity network users has reached millions. According to the statistics on Tor and I2P official websites, about 3 million users use Tor repeaters to connect to the Internet, and 36,000 I2P network routing nodes are running.

However, while the anonymity network brings privacy protection and greater freedom to ordinary users, it also hides the identity information of malicious users. Malicious users use anonymity networks to evade network supervision and engage

in various network violations, which brings new challenges to network supervision. Some studies [1], [2] analyzed the structure and content of Tor and I2P, which pointed out that there is a large amount of illegal content in anonymity networks. Thus, it is necessary to adopt effective methods to supervise anonymity networks.

With the public release of anonymity network datasets ISCX2016 and Anon17 [3], [4], the classification of anonymity network traffic has also become a hot research field. To identify anonymity network services in massive and complex traffic data, many studies have used machine learning (ML) algorithms such as Naive Bayes (NB), Random Forest (RF), k-Nearest Neighbor (KNN) as classifiers and achieved good performance [3]–[6]. However, the ML-based methods have some constraints, which have limited feature extraction capabilities and require manual feature selection. Deep learning (DL) algorithms have better feature extraction capabilities and have been widely used in traffic classification tasks recently. For instance, the Convolutional Neural Network (CNN) [7]–[9] and the Long Short-Term Memory (LSTM) [10] have better performance in classifying network traffic, and the accuracy of classifying application traffic is more than 80%. However, most of the aforementioned methods only consider the characteristics of a single flow, i.e., do not consider the relationship between the flows. Since each flow is regarded as an isolated individual and is sequentially input into the neural network for feature extraction, the relationship between the flows is not effectively utilized. According to the temporal and spatial correlation of the traffic, there are usually some correlations among multiple continuous flows generated in a short period of time. Although some works [11]–[14] regarded continuous multiple flows as a flow sequence for feature extraction, they did not distinguish the importance of different flows. Real-world traffic is often mixed with unimportant or irrelevant flows, which will adversely affect the identification performance of such methods. To achieve an anonymity network traffic classifier with excellent performance, we design our method from the following three key points:

- **Use flow sequence for classification:** The flow sequence is composed of continuous flows in time, which obviously contains more information than a single flow.
- **Distinguish the importance of flows:** To reduce the noise brought by irrelevant flows and pay more attention

¹<https://www.torproject.org/>

²<https://geti2p.net/>

³<https://anonymouse-proxy-servers.net/>

to important flows, it is necessary to distinguish the importance of different flows in the flow sequence.

- **Select the optimal feature combination:** Dozens of features are parsed from the original traffic, and feature selection can avoid the insignificant features from reducing the performance and efficiency of the classifier.

Based on the aforementioned first two key design points, we propose a novel flow-based anonymity network traffic classifier with multi-head attention mechanism, called FLOW TRANSFORMER. Based on the third point, we adopt the RF-based method to select the optimal feature combination. In our proposed method, the flow sequence is the input feature, which is composed of multiple continuous flows within a certain time range. We represent the input flow sequence as three vectors of query (Q), key (K), and value (V), where Q and K are used to extract representative features of the flow, and V is the information of the flow itself. Through an effective multi-head attention mechanism (i.e. calculate $Q \cdot K$), FLOW TRANSFORMER can distinguish the importance of different flows in each flow sequence. The major contributions of the proposed work are three-fold:

- We propose an anonymity network traffic classifier with multi-head attention mechanism, which can pay more attention to important flows and extract flow sequence features more effectively. To the best of our knowledge, this is the first work to use the multi-head attention mechanism to distinguish the importance of different flows in each flow sequence.
- Insignificant features in the flow can be characterized as noise under certain conditions, which can have a very adverse effect on the identification result. In addition, too many features will bring more model parameters, which leads to slower training and classification. Thus, in order to improve the performance and efficiency of the classifier, we adopt the RF-based method to select the optimal feature combination.
- To comprehensively evaluate the effectiveness and generality of FLOW TRANSFORMER, we conducted experiments on SJTU-AN21 and ISCXVPN2016 datasets. Experiments show that FLOW TRANSFORMER can achieve better performance than other classifiers.

II. RELATED WORK

The task of network traffic classification has been continuously concerned by researchers, but not all network traffic analysis methods are suitable for anonymity network traffic. In this section, we will introduce the existing research status of anonymity network or other encrypted network traffic classification methods.

The anonymity network uses encryption for communication and transmission. Thus, methods such as deep packet inspection (DPI) [15] that requires analysis of packet information cannot classify encrypted traffic. Recently, researchers identify anonymity network traffic by analyzing behavior and statistical characteristics, which are mainly based on ML or DL algorithms. Lashkari *et al.* [3] proposed a ML-based method to

detect Tor traffic through time features. The experimental results show that only 10 time features are needed to effectively detect different applications in Tor traffic. Elike *et al.* [16] proposed a Tor network off-line traffic classification system using SVM classifier to improve the performance of traffic classification. Montieri *et al.* [17] proposed the classification of different anonymous services at different granularity levels, which uses five ML algorithms to classify anonymity services with a maximum accuracy of 85.8%.

The above results show that these ML-based methods can achieve good classification accuracy in the face of uncomplicated network traffic. However, they cannot cope with increasingly complex network traffic. Due to better robustness and feature extraction capabilities, DL-based method has become the most ideal traffic classification method. Wang *et al.* [7], [8] transformed the problem of traffic classification into an end-to-end image classification problem, and proposed an encrypted traffic classification method based on 2D-CNN, which achieved good accuracy by classifying traffic images. Chen *et al.* [18] proposed a capsule neural network method to identify encrypted traffic and achieved higher classification accuracy. Lotfollahi *et al.* [19] proposed a traffic classification model based on CNN combined with stacked autoencoder (SAE), which achieves more effective classification by reconstructing the characteristics of raw traffic data.

The aforementioned methods automatically extract the characteristics of flows, but ignore the temporal-spatial correlation of the traffic or the different importance of the flow. Thus, we explore the problem of anonymity network traffic classification from the perspective of flow sequence. Our goal is to learn the importance of different flows and apply it for feature extraction.

III. PRELIMINARIES

In this section, we introduce some background of anonymity network traffic classifiers, including the operating mechanism of anonymity networks and the overview of *Transformer*.

A. Anonymity Network

The three most popular anonymous networks are Tor, I2P and JonDonym. Tor is an anonymity communication network with the largest number of users. It is based on TCP and uses a multi-hop mechanism to build communication links. In an established communication link, between the ingress node and the egress node, the Tor network will randomly select more than three relay nodes from the directory server. Each relay node only knows its previous and next nodes, the selection of nodes is random, and the link is constantly changing. At the same time, the data in the Tor network link is hidden in multiple encryption layers, and all communications are encrypted layer by layer. I2P is an improved anonymity network based on Tor, which removes the central node and adopts a completely distributed structure to improve the anonymity and stability of the network. It uses a multi-level encrypted tunnel mechanism to hide the identity information and communication relationship of the communicating parties.

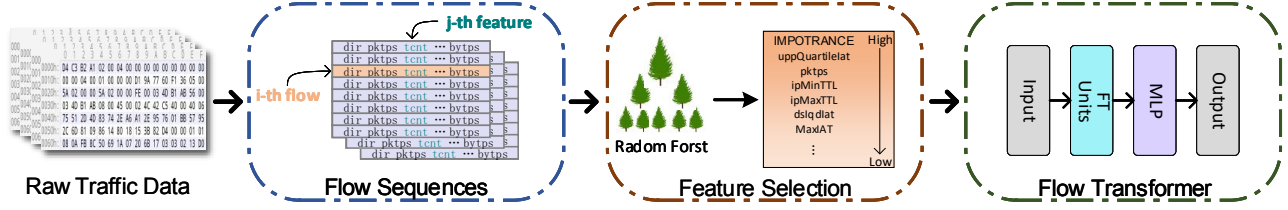


Fig. 1. Overview of proposed method for anonymity network traffic classification. The flow sequence represents a set of continuous multiple flows. The Flow Transformer Unit (FT Unit) is a flow sequence feature extraction unit with multi-head attention mechanism.

JonDonym provides users with anonymous services through multiple hybrid cascades. There is a set of two or three encrypted hybrid servers in each hybrid cascade. Unlike Tor, the link of the JonDonym network is fixed. The user can select the link for data transmission, but cannot change the mix of different nodes on the link.

Due to the application of encryption technology, the classification of anonymous network traffic is more challenging.

B. Transformer with Multi-head Attention Mechanism

The *Transformer* was first proposed by Vaswani *et al.* [20] in 2017 and has achieved great success in the field of NLP [21] and CV [22]. The *Transformer* employs an encoder-decoder structure. The self-attention layer is one of the most important structures of the *Transformer* network, which uses residual connections followed by layer normalization. Therefore, the output of self-attention layer is $LayerNorm(x + Sublayer(x))$.

The self-attention layer employs h attention heads. The results of different heads are concatenated together and a linear transformation is used. As for attention head i , first calculate Q_i, K_i, V_i by using linear transformation, representing Query Matrix, Key Matrix and Value Matrix respectively.

$$Q_i = xW_i^Q, K_i = xW_i^K, V_i = xW_i^V \quad (1)$$

$W_i^Q, W_i^K, W_i^V \in \mathbb{R}^{d_x \times d_k}$ are learnable parameters. Then the output of a single attention head is computed:

$$Attention(Q_i, K_i, V_i) = softmax(\frac{Q_i K_i^T}{\sqrt{d_k}}) V_i \quad (2)$$

Matrix calculation can be parallelized and reduce calculation time. The essence is to use the similarity between the query vector and the key vector as the weight, and perform a weighted summation on all value vectors.

Multi-head attention allows the model to jointly attend to information from different head:

$$MultiHead(x) = Concat(head_1, \dots, head_n) W^O \quad (3)$$

where $head_i = Attention(Q_i, K_i, V_i)$ and $W^O \in \mathbb{R}^{hd_k \times d_x}$ is used for linear transformation. Through multi-headed attention, information in different spaces can be extracted effectively, and richer feature information can be captured.

The *Transformer* network with multi-head attention mechanism can assign different attention weights to flows in the

flow sequence, which realizes reasonable and effective feature extraction of the flow sequence.

IV. SCHEME OF ANONYMITY NETWORK TRAFFIC CLASSIFICATION

In this section, we introduce the scheme of anonymity network traffic classification. After data preprocessing and feature selection, FLOW TRANSFORMER is adopted to realize the anonymity services classification. The overview of the proposed method is shown in Fig. 1 and the process of our approach is summarized in **Algorithm 1**.

Algorithm 1 Pseudocode of the Proposed Method

Input: raw training dataset D_r , total epoch times n

Output: FLOW TRANSFORMER for anonymity network traffic classification

- 1: **procedure** Data Preprocessing (D_r)
- 2: Compute the standardization result according to (4)
- 3: Take continuous multiple flows as flow sequences
- 4: Use RF-based method to select feature combination
- 5: $D_p \leftarrow$ New training dataset after preprocessing
- 6: **return** D_p
- 7: **procedure** Flow Transformer Model (D_p)
- 8: **while** $i \leq n$ **do**
- 9: Load the proposed network
- 10: Input D_p into the network for training
- 11: Update the model according to cross entropy loss
- 12: Save *Flow Transformer_i* model
- 13: **end while**
- 14: Save *Flow Transformer_n* as the final model

A. Data Preprocessing

Before the training dataset and the testing dataset are applied to the FLOW TRANSFORMER model, the data need to be preprocessed. This involves two steps.

1) *Standardization*: Due to the large range of values with different characteristics in each flow, directly using raw data for training will weaken the effect of characteristics with lower values. In addition, the convergence speed of the model will be very slow. Therefore, it is necessary to use the standardization algorithm to ensure the reliability of data. The algorithm is adopted to standardization the data as follows:

$$z = \frac{x - \mu}{\sigma} \quad (4)$$

where μ represents the mean of the original data, and σ the standard deviation of the original data.

2) *Flow Sequences Generation*: The flow sequence is composed of multiple continuous flows. As these flows are continuous in time, they can reflect user behavior during this period. Some studies have shown that traffic classification by the flow sequence has better performance than by only one flow [11]–[14]. Thus, in our method, we take 8 continuous flows as a sequence.

B. RF-based Method for Feature Selection

Feature selection is to remove insignificant features in the generated flows, which can make the generated results better in line with effectiveness and timeliness. In terms of effectiveness, insignificant features in the flow can be characterized as noise under certain conditions, which can have a very adverse effect on the identification result. In terms of timeliness, the removal of these features can reduce the calculation of statistics in flow generation process and speed up flow generation. In addition, low-dimensional features will also reduce the complexity and classification time of the feature extraction network.

RF algorithm is robust and easy to use, making it one of the most popular ML algorithms for classification tasks. Moreover, RF algorithm can be employed to calculate the contribution of each feature, which is effective in feature selection. A total of 84 initial features are extracted from the raw traffic data. We use RF-based method to evaluate the importance of these 84 features and perform feature selection. The calculation of the j -th feature contribution based on the RF method is divided into four steps. First, randomly sample the original training set D as the in-bag data D_{in} , and the unselected data as the out-of-bag data D_{out} ; Next, use the in-bag data D_{in} to train the RF model. For the i -th decision tree in RF model, use the corresponding out-of-bag data D_{out} to calculate the data error, denoted as Err_{i1} ; Then, randomly add noise to all the sample features, and calculate the error of D_{out} again, denoted as Err_{i2} ; Finally, the contribution of the j -th feature is calculated according to the following formula:

$$c_j = \sum_{i=1}^N \frac{Err_{i2} - Err_{i1}}{N} \quad (5)$$

where N represents the number of trees in the RF model. It can be seen that if the accuracy of D_{out} drops significantly after random noise is added, it means that this feature has a great influence on the prediction results of the sample, i.e., the importance is relatively high. Based on this method, we sort the importance of the initial features and optimize the feature combination.

C. Flow Transformer Classifier

Different from other methods, FLOW TRANSFORMER classifier can assign different attention weights to different flows in the flow sequence through the multi-head attention mechanism. The multi-head attention mechanism strengthens the information interaction between related flows, and can achieve

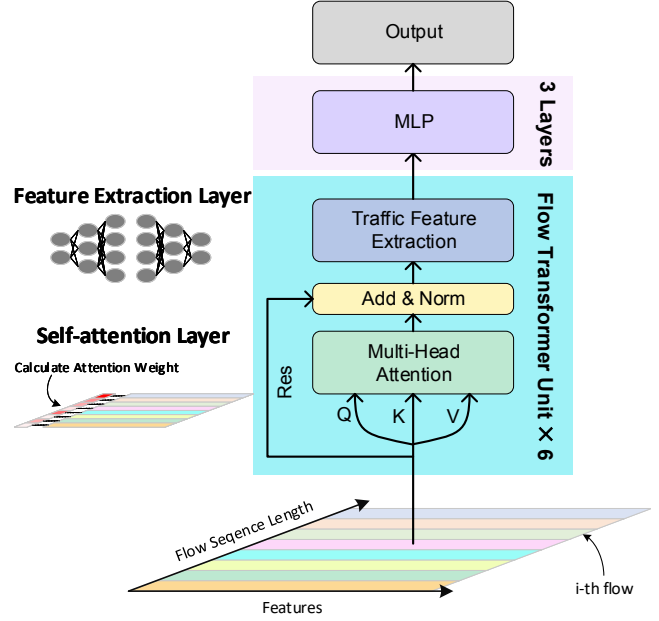


Fig. 2. The structure of FLOW TRANSFORMER.

reasonable and effective feature extraction of the flow sequence.

The input of FLOW TRANSFORMER is a flow sequence consisting of 8 continuous flows after feature selection. Continuous flows can be regarded as a sequence input. The model structure of FLOW TRANSFORMER is shown in Fig. 2. It consists of 6 Flow Transformer units and 3 fully connected layers. Two key sub-layers form a Flow Transformer unit, namely self-attention layer and feature extraction layer.

The self-attention layer is mainly used to distinguish the importance of different flows in each flow sequence. Query Matrix, Key Matrix and Value Matrix are calculated by linear transformations, denoted as Q , K and V . Here, Q and K represent key information of the flows, while V represents the actual information of the flows. The higher the similarity between Q and K , the higher the correlation between the flow corresponding to Q and the flow corresponding to K , and therefore the higher the weight when V is weighted and summed. The multi-head attention mechanism strengthens the information interaction between related flows and effectively improves the feature extraction effect of the model on the flow sequence.

The feature extraction layer can strengthen the expressive ability of the output. It is able to provide non-linear transformation functions that self-attention layer does not have. It first uses a fully connected layer to map features to a larger dimensional feature space, then uses ReLU for nonlinear screening, and finally uses another fully connected layer to restore the features to the original dimension.

Each sub-layer uses the residual structure, and added layer normalization at the end. The residual structure can solve the

degradation problem that occurs as the depth of the network increases, while layer normalization can ensure the stability of the data distribution and accelerate the convergence speed of the model.

Finally, the output of Flow Transformer units is flattened and a MLP with three fully connected layers is used for classification. The final layer has 10 or 7 neurons, which is equal to the number of network service categories.

V. EVALUATION

In this section, we present and discuss our experimental results. Especially, we answer the following three research questions:

- **RQ1:** How effective is the RF-based method in feature selection?
- **RQ2:** How well does FLOW TRANSFORMER work on real-world anonymity network traffic, identifying different anonymity services?
- **RQ3:** How well does FLOW TRANSFORMER perform comparing to the state of art of traffic classification methods?

A. Dataset for Evaluation

The previous anonymity network datasets (i.e. ISCX-Tor2016 and Anon17) have become invalid due to the update of the anonymity network version. The Tor network has been significantly changed in version 0.4.1.5 in 2018, which adds circuit-level padding and SENDME units to prevent deanonymization through traffic analysis. The I2P network has enabled the new protocol NTCP2 in version 0.9.36 in 2019. The NTCP2 uses the Noise protocol framework to improve the ability to resist DPI attacks. However, the previous two anonymity network datasets were released before 2017 [3], [4]. To make the classifier applicable to the current anonymity network traffic analysis, in the latest version of the three most popular anonymity networks (i.e. Tor, I2P, JonDonym), we collected the traffic data generated by 10 anonymity services. This latest anonymity network traffic dataset SJTU-AN21⁴ is publicly available for researchers.

In addition, encrypted traffic has some similarities with anonymous network traffic, so the public encrypted traffic dataset ISCXVPN2016 is used to evaluate the generality of the proposed method.

Details of these two datasets are summarized in TABLE I. In the SJTU-AN21 dataset, Eepsites, IRC, Snark, and Video are the four main anonymity services in the I2P network, and Bittorrent, Chat, FTP, Streaming, and Browsing are the five main anonymity services in the Tor network.

B. Experiment Setup

We design several experiments to evaluate the efficacy of our method for classifying anonymity network traffic. To answer RQ1, we evaluate the performance of the RF-based feature selection method and the traditional PCA method on different number of features, and determine the optimal feature

TABLE I
SUMMARY OF THE DATASETS USED FOR EVALUATION.

Dataset	Category	Train dataset	Test dataset
SJTU-AN21	Eepsites	1,825	1,197
	IRC	3,009	484
	Snark	7,284	1,784
	Video	12,577	581
	JonDonym	1,254	967
	Bittorrent	198	76
	Chat	624	150
	FTP	364	184
	Streaming	949	598
	Browsing	1,130	958
	Total	29,214	6,979
ISCXVPN2016	Browsing	3,646	912
	FTP	4,446	1,112
	P2P	736	184
	VoIP	8,464	2,117
	Email	689	173
	Streaming	1,523	381
	Chat	3,209	803
	Total	22,713	5,682

combination for subsequent experiments. To answer RQ2, we analyze the training process of FLOW TRANSFORMER, and discuss the confusion matrix of the classification results on the test dataset. To answer RQ3, we compare the classification performance of FLOW TRANSFORMER and other state of art of traffic classification methods on the test dataset. It should be noted that both SJTU-AN21 and ISCXVPN2016 datasets are used to conduct the above evaluation experiments. To make the experimental evaluation clearer, we introduce the experimental environment and evaluation metrics as follows.

1) *Experimental Environment:* Our anonymity network traffic is collected by configuring the mirror port of the switch and using the real-time traffic capture tool *tcpdump*. The raw traffic data of SJTU-AN21 and ISCXVPN2016 datasets are all pcap files, which need to calculate statistical features to enrich feature information. Thus, we use *Tranalyzer*, a lightweight packet analyzer, to calculate the statistical characteristics of the raw traffic data. All the evaluations are conducted in Python 3.7 with the PyTorch framework of version 1.8.1 and running on the PC with Intel® Core™ i7-10700@2.90 GHz, 16 GB RAM, a 512G SSD, and an NVIDIA GeForce RTX3060 12GB.

2) *Evaluation Metrics:* The related evaluation metrics used in the evaluation are as follows. There are two situations in which the classifier performs a correct classification. One is the true positive T_p decision, which correctly classifies the traffic corresponding to the service. The other is true negative T_n decision, i.e. the traffic of other services is correctly classified.

⁴<https://github.com/iZRJ/The-SJTU-AN21-Dataset>

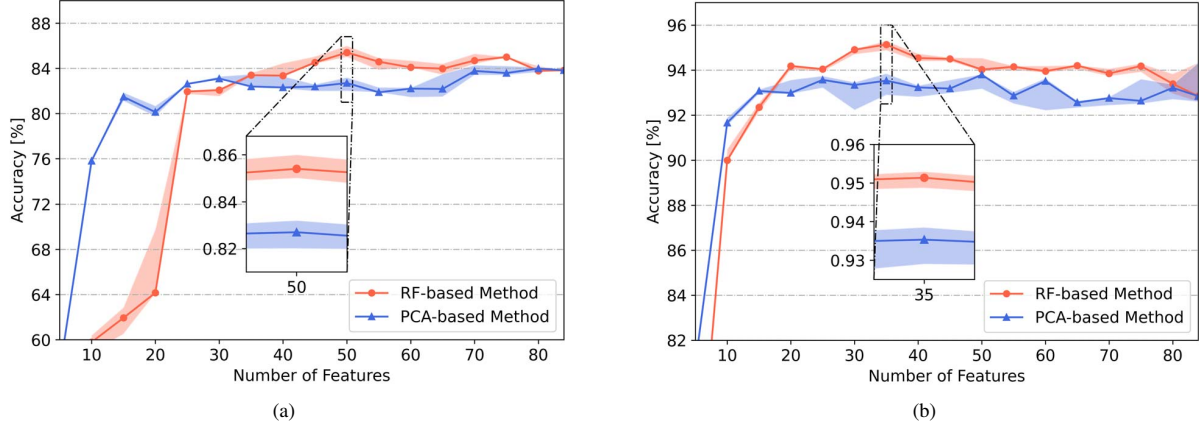


Fig. 3. Classification accuracy with different numbers of features on (a) SJTU-AN21 dataset, (b) ISCXVPN2016 dataset. The shadow in the figure indicates the variation range of the top-3 accuracy, and the connection point represents the second highest accuracy using different number of features.

The corresponding erroneous output of the classifier is false positive F_p decision and false negative F_n decision. Based on the above definition, Recall, Precision and F_n can be obtained:

$$Recall = \frac{T_p}{T_p + F_n} \quad (6)$$

$$Precision = \frac{T_p}{T_p + F_p} \quad (7)$$

$$F_n = \frac{(n^2 + 1)Precision \cdot Recall}{(n^2 Precision + Recall)} \quad (8)$$

where n is a penalty factor to provide more weight to recall, and we choose F_1 in this paper.

C. Efficacy of RF-based Feature Selection Method (to RQ1)

As we mentioned before, insignificant features can adversely affect the performance of the classifier. At the same time, more features mean longer training time and prediction time. Thus, in order to obtain the optimal feature combination, we first conduct the evaluation of feature selection.

In this stage, we choose another representative feature selection method, principal component analysis (PCA) algorithm to compare the performance of feature selection. As an unsupervised method, the PCA algorithm is widely used for data analysis. According to the RF-based and the PCA-based feature selection methods, we investigate the performance of the classifier with different number of features (i.e., the top J features in the rank). The experimental results are shown in Fig. 3. To avoid the contingency of the classifier performance, we take top-3 accuracy during the training process. The shadow in the figure indicates the variation range of the top-3 accuracy, and the connection point represents the second highest accuracy using different number of features. Obviously, the RF-based method for feature selection has better effectiveness and stability.

Moreover, we can see that if the number of features is less than 20, satisfactory performance cannot be achieved on either the SJTU-AN21 dataset or the ISCXVPN2016 dataset. On the SJTU-AN21 dataset, when the number of features is 50, the FLOW TRANSFORMER reaches the highest accuracy of 85.29%. On ISCXVPN2016, the highest accuracy of the classifier is 95.29% when the number of features is 35. Due to the use of more complex and up-to-date encryption technology, the anonymity network requires more features to effectively classify traffic than encrypted networks. We use the above optimal feature combination on these two datasets to conduct subsequent experiments.

D. Performance of Flow Transformer Classifier (to RQ2)

The train accuracy and loss using the FLOW TRANSFORMER classifier for SJTU-AN21 and ISCXVPN2016 datasets are shown in Fig. 4. The loss of FLOW TRANSFORMER classifier decreases rapidly and eventually converge to near zero, which indicates that the classifier has learned the feature information of the training dataset very well. Benefiting from the excellent feature extraction ability of the FLOW TRANSFORMER for network traffic, its final training accuracy is very high.

To comprehensively evaluate the classification performance of FLOW TRANSFORMER, we analyze the confusion matrix of the classification results on the test datasets. As shown in Fig. 5, our classifier can effectively classify the traffic generated by different services. In Fig. 5(a), the traffic of the three types of anonymity networks (i.e., I2P, Tor, JonDoNym) are correctly distinguished. Some of the *Chat* services of the Tor network are misclassified as the *Streaming* service. Since some files are transferred during the *Chat* service, the corresponding traffic is very similar to *Streaming* service. In Fig. 5(b), we can see that the classifier has some classification errors between the *FTP* and *Chat* service traffic. By referring to the official description of the ISCXVPN2016 dataset, we found that both

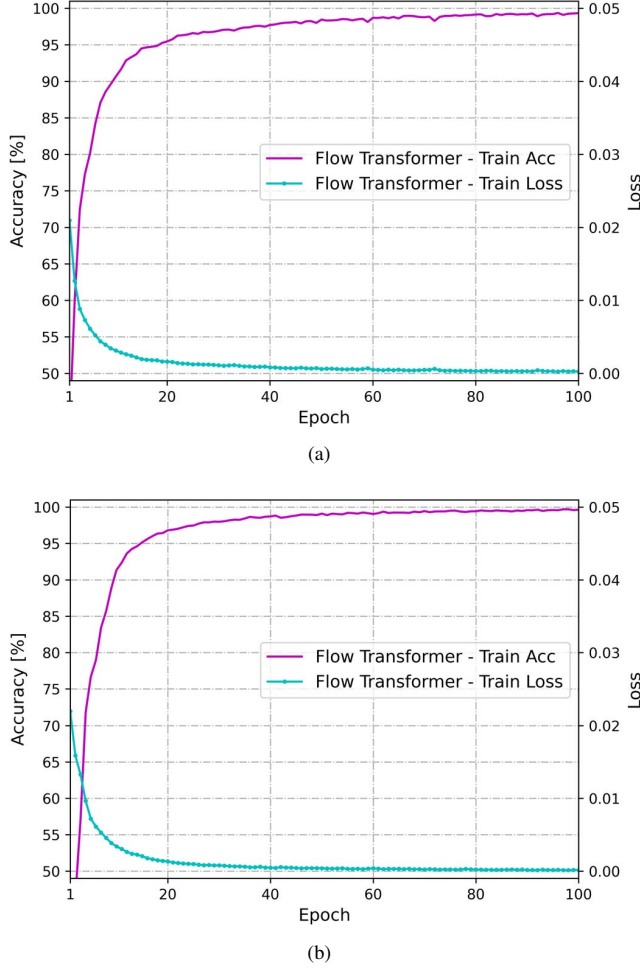


Fig. 4. Train accuracy and loss of FLOW TRANSFORMER classifier on (a) SJTU-AN21 dataset, (b) ISCXVPN2016 dataset.

services contain the traffic generated by the Skype application, which is the main cause of misclassification.

E. Comparison with Other Methods (to RQ3)

We compare our with the following nine baseline methods on SJTU-AN21 and ISCXVPN2016. The ML models (i.e. **NB**, **RF**, **SVM**, and **C4.5**) are implemented through the popular data mining tool *Weka*. **CNN**, **LSTM**, and **LSTM+Attn** models all use the structure similar to FLOW TRANSFORMER (i.e. the same number of layers and input/output size), and use flow sequences as input for classification. In addition, **LSTM+Attn** uses LSTM combined with a self-attention mechanism to pay attention to important flows in the flow sequence. **2D-CNN** is proposed by Wang *et al.* [7] that directly reads traffic pcap files and converts them into images for classification. **3D-CNN** is an improved model of the 2D-CNN proposed by Zhang *et al.* [9], which uses multiple channels to enrich feature information during image conversion.

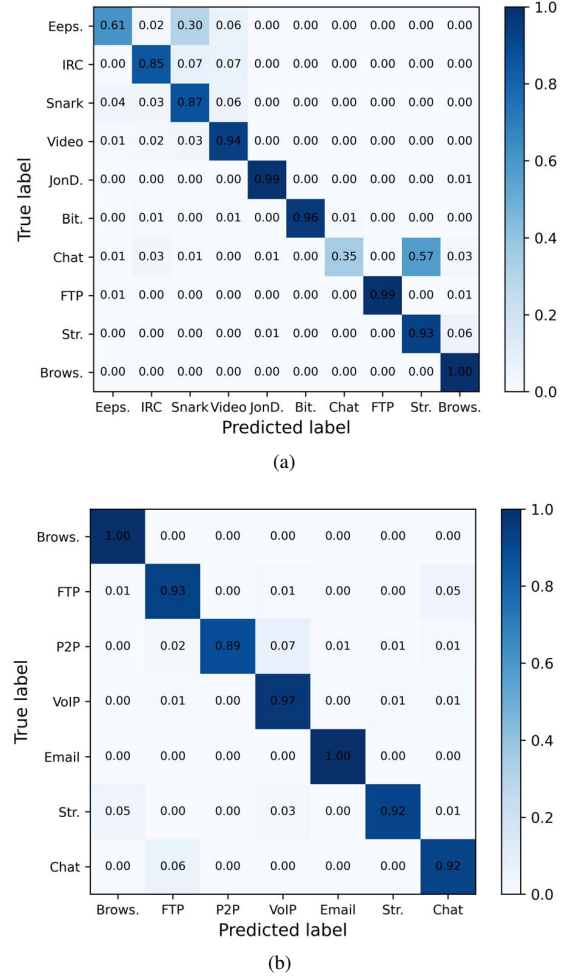


Fig. 5. Performance confusion matrices of FLOW TRANSFORMER classifier on (a) SJTU-AN21 dataset, (b) ISCXVPN2016 dataset.

It can be seen from TABLE II that our FLOW TRANSFORMER achieves the best performance in both two datasets in terms of all evaluation metrics. We can observe that the classification results of traditional ML-based methods are usually not ideal, demonstrating the limited ability of these methods to classify complex network traffic. The classification performance of the 2D-CNN and 3D-CNN models (i.e., directly reading pcap files without calculating the statistical features) is very limited. CNN and LSTM models use statistical features and the flow sequence to achieve significant performance improvements. However, due to the lack of mining the inherent information of the flow sequence, it is still unable to achieve high accuracy. We introduce a self-attention mechanism on the LSTM model, and the detection performance has been improved by about 3%, which shows that the attention mechanism is very effective for the flow sequence. Our FLOW TRANSFORMER uses the multi-head attention mechanism to distinguish the importance of flows, and the feature extraction layer to fully extract flow sequence features, which greatly

TABLE II
THE PERFORMANCE COMPARISON WITH OTHER METHODS.

Model	SJTU-AN21			ISCXVPN2016		
	Acc.	Prec.	F_1	Acc.	Prec.	F_1
NB	34.0%	34.1%	27.0%	54.7%	68.1%	57.2%
RF	67.9%	73.7%	67.1%	85.8%	88.0%	85.8%
SVM	46.8%	58.0%	42.8%	71.8%	75.2%	72.0%
C4.5	48.4%	55.8%	48.7%	78.5%	78.0%	76.2%
CNN	78.3%	78.4%	77.0%	90.2%	90.7%	90.4%
LSTM	79.1%	80.2%	77.9%	90.2%	90.8%	90.3%
2D-CNN	65.0%	70.8%	66.9%	86.2%	86.7%	86.1%
3D-CNN	71.9%	75.6%	73.2%	87.6%	88.1%	87.9%
LSTM+Attn	81.5%	82.1%	81.1%	92.3%	92.9%	92.3%
Ours	86.0%	86.8%	85.5%	95.2%	95.3%	95.2%

improves the classification performance of anonymity network and encrypted network traffic. Further, we also try to add CNN, LSTM or autoencoder network to the existing network structure of FLOW TRANSFORMER for feature extraction. Experimental results show that these methods will not improve the performance of the classifier, and we do not want our FLOW TRANSFORMER to be a highly complex model. Thus, the multi-head attention mechanism and the feature extraction layer of FLOW TRANSFORMER are all we need.

VI. CONCLUSION

In this paper, a novel flow-based traffic classifier called FLOW TRANSFORMER is proposed and successfully applied to classifying anonymity network and encrypted network traffic. The model uses the multi-head attention mechanism to effectively distinguish the importance of the flow in the flow sequence, and the feature extraction layer to fully extract the flow sequence features. Besides, to avoid the insignificant features from reducing the performance and efficiency of the classifier, we use the RF-based feature selection method to select the optimal feature combination. Experiments on two real-world traffic datasets show that the classifying accuracy of the proposed model is superior to existing models. In the future, we will consider combining character features in traffic to further improve classification performance. Since the FLOW TRANSFORMER is a general classifier for network traffic, we can also apply it to other traffic analysis tasks, such as network intrusion detection.

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